Individual Reflection

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I completed this coursework as an individual submission, so I was responsible for all stages of the work, including implementation, experimentation, analysis, report writing, and preparation of the final submission files.

For Part 1, my main role was to implement the required CIFAR-10 image classification architecture from the assignment brief. The most important task was translating the written description of the intermediate block into working PyTorch code. I implemented the independent convolutional paths, the channel-average weighting mechanism, the output block, the full model, and the training and evaluation pipeline. I also created the CIFAR-10 DataLoaders and generated the required training loss and accuracy plots.

After completing the initial model, I worked on improving the result while keeping the architecture close to the required design. I increased the number of channels and convolutional paths per block, used data augmentation, AdamW, weight decay, and a cosine learning-rate scheduler. This improved the best testing accuracy from 84.76% to 87.81%. This part helped me understand the importance of keeping architectural changes within the assignment constraints rather than simply replacing the model with a standard CNN.

For Part 2, I designed and implemented a Fashion-MNIST ablation study. I compared four CNN variants: a baseline CNN, a BatchNorm CNN, a BatchNorm CNN without dropout, and a BatchNorm CNN trained with data augmentation. I implemented the model variants, training loop, evaluation code, results tables, and plots. The aim was to test how batch normalization, dropout, and augmentation affected generalisation performance under the same training budget. The strongest result came from the BatchNorm CNN, which achieved a best testing accuracy of 92.50%.

I also wrote the final report. This included explaining the model architecture, justifying the training setup, describing the experimental protocol, interpreting the results, discussing limitations, and adding relevant references. I prepared the final notebooks and checked that the outputs, figures, tables, and results matched the report.

Overall, this coursework developed my ability to implement a neural network from a high-level specification, run controlled experiments, and explain results clearly. The main challenge was balancing performance improvement with the requirement to preserve the custom architecture in Part 1. For Part 2, the main lesson was that a stronger experiment comes from comparing controlled model variants rather than only reporting one model’s performance.